

Designing Sensory Metaphors in Virtual Reality for understanding Building Acoustic

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Abstract—This paper presents the design and preliminary evaluation of an immersive virtual reality (VR) learning environment dedicated to the understanding of acoustic phenomena in buildings through sensory metaphors. Rather than reproducing acoustics with strict physical realism, the proposed approach translates sound propagation, reflection, transmission, attenuation, and reverberation into perceptible visual and haptic representations designed to support intuitive interpretation. This work contributes both a VR educational framework for building acoustics training and a methodological approach for designing sensory metaphors dedicated to invisible physical phenomena. A preliminary experimental study was conducted with high school students using pre/post knowledge evaluation, usability assessment, and user experience analysis. (*Abstract*)

Keywords—virtual reality, sensory metaphors, learning, building, acoustic, user experience

I. INTRODUCTION

A. Context

Virtual reality (VR) is increasingly used in professional building education, and professional simulation because of its ability to provide immersive, interactive, and experiential learning situations. In the building and architectural domains, immersive technologies are now employed for construction planning and renovation workflows [1], immersive BIM environments [2, 8], environmental simulation [4], architectural acoustics [5], and professional education scenarios [6]. These systems allow learners and professionals to interact directly with complex spatial and physical situations that are difficult, expensive, or impossible to reproduce in traditional educational contexts.

In professional building education, immersive environments offer a number of educational benefits. They enable learners to explore architectural spaces on a real-world scale, manipulate environmental parameters in real time, and immediately observe the consequences of physical changes. Unlike traditional classroom methods, which rely on diagrams, equations or standard tables, virtual reality environments can transform

abstract physical processes into concrete, interactive sensory experiences. Such approaches are particularly relevant for invisible phenomena, including sound propagation, reverberation, air flows, thermal comfort and environmental interactions. Recent research has also explored sensory and multisensory representations capable of making these phenomena perceptible through visual, auditory and haptic channels [7, 8, 9, 10, 11, 12]. These immersive representations promote experiential learning by enabling learners to directly perceive spatial and physical relationships that would otherwise be inaccessible in conventional teaching situations.

Furthermore, teaching building physics remains, however, a major challenge in professional building education, particularly in the field of acoustics, where most physical phenomena are difficult to grasp. Students are expected not only to apply technical standards and calculation methods, but also to understand the physical mechanisms governing acoustic comfort within architectural spaces. However, many learners struggle to make meaningful connections between theoretical calculations, regulatory constraints, material properties and the dynamic behaviour of sound within buildings. Concepts such as sound propagation, reverberation, attenuation or transmission between adjacent rooms are generally presented using formulas, simplified diagrams and standardised tables, which offer only limited assistance in understanding these physical mechanisms.

In addition, in high schools, practical application is also limited by the cost of materials, the available infrastructure and technical constraints. Students cannot easily alter the composition of walls, insulation materials or the geometry of rooms to compare their acoustic effects in real time. Similarly, to understand phenomena such as reverberation or sound transmission, students often have to mentally reconstruct dynamic spatial interactions from static representations. This gap between theoretical representations and perceptual understanding constitutes a major pedagogical constraint in the current teaching of building acoustics.

Virtual reality therefore appears as a promising medium for experiential and perceptual learning in this context. By enabling

learners to explore immersive architectural environments and directly interact with physical simulations, VR can support the understanding of invisible phenomena through embodied and situated experiences. The present work specifically investigates the use of sensory metaphors to render acoustic phenomena perceptible in immersive environments. The proposed system relies on independent visual and haptic metaphors sharing common acoustic parameters derived from simplified building physics models. Rather than aiming for complete physical realism, the objective is to provide interpretable immersive representations capable of supporting operational understanding aligned with professional learning objectives.

The proposed approach aims to contribute both to immersive learning research and to the development of pedagogical tools adapted to the specific challenges of professional building education.

That is why, following the review of the current state of the art below, the “Method” section will explain the approach taken. Indeed, to evaluate this approach, an experimental protocol was conducted with vocational high school students in the building field. An evaluation of acoustic physics knowledge was carried out before and after the VR experiment. Participants were divided into three experimental conditions, all of which took place in virtual reality. Finally, the “Results and Discussion” section will present the results of assessments of cognitive load (RTLX) and usability (SUS), as well as responses to open-ended questions, which were carried out at the end of the experimental protocol, before concluding in the final section.

II. STATE OF ART

In order to design a virtual reality learning environment dedicated to vocational training in the construction sector, this state-of-the-art review critically analyses existing immersive systems in light of the objectives outlined in the previous section: to improve understanding of invisible acoustic phenomena, to support novice learners, and to preserve perceptual intelligibility within immersive environments. Rather than evaluating this work solely in terms of simulation fidelity or technical realism, the analysis focuses on how immersive representations transform physical phenomena into perceptible and interpretable experiences. Particular attention is therefore paid to sensory metaphors, perceptual correspondences, multimodal interactions, cognitive accessibility and pedagogical transferability.

• Immersive metaphors of acoustic phenomena

Several studies in architectural acoustics [7, 8, 9, 10] seek to make acoustic phenomena perceptible and interpretable through immersive representations and perceptual metaphors.

The work of Deines et al. [7, 8] is representative of this approach. Their virtual reality systems visualise acoustic propagation using: acoustic rays, animated wavefronts, reverberation fields, spatial trajectories, and colour-coded propagation intensities (see Figure 1 and 2). These representations transform acoustic calculations into structures that can be observed in space. In addition to hearing the sound, the user can visually observe how it interacts with the geometry, surfaces and obstacles.

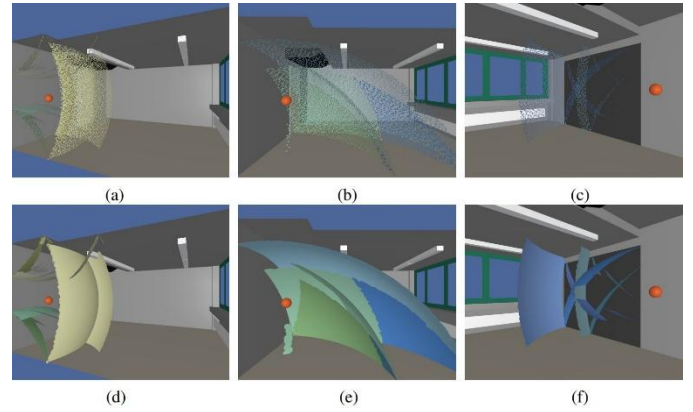


Fig. 1. Wave fronts reflected from walls (a+d), floor (b+e), and projection wall (c+f) using smaller (a,b,c) and greater (d,e,f) radius [7]

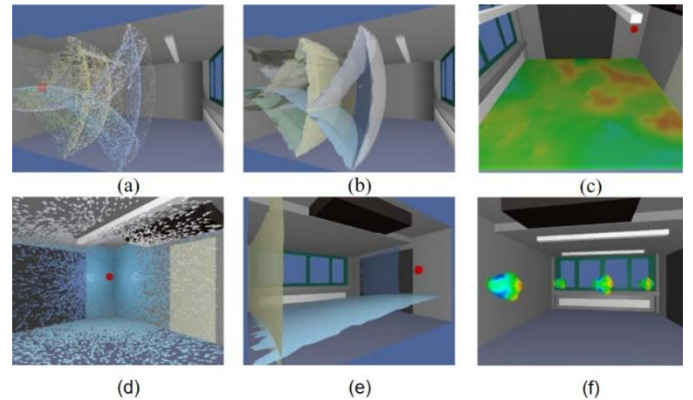


Fig. 2. Visualization of the simulation results calculated using the phonon tracing [8]

Similarly, Astheimer [8] proposed one of the first immersive acoustic metaphors through the principle ‘What you see is what you hear’, where visual spatial cues are synchronised with acoustic propagation behaviours. This approach introduces a perceptual correspondence between visual movement and auditory perception, with the aim of improving the intuitive understanding of spatial acoustics.

It is important to note that some studies explicitly encode acoustic performance using visual symbolic mappings (explained below). Armin Taghipour, Tessa Sievers and Kurt Eggenschwiler [10], for example, visually represent the acoustic performance of building façades through colour coding in immersive urban environments (see Figure 3). Here, colours become symbolic metaphors for sound attenuation and comfort levels. This is particularly significant, as it transforms abstract acoustic measurements into directly interpretable spatial information.

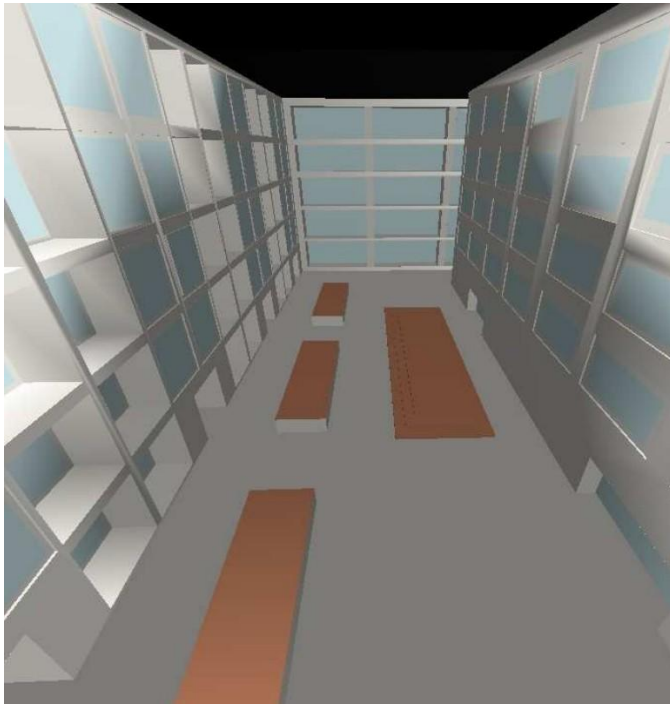


Fig. 3. Absorption coefficient of the materials is coded by their colors [10]

However, despite their sophistication, these visualisations are still intended for experts, architects, acoustic engineers and simulation specialists, rather than high school students, which is not in line with the objectives. Most systems assume that users are already familiar with acoustic terminology, propagation principles, simulation parameters and interpretation conventions.

- **Immersive simulation of acoustic phenomena**

Firstly, Michael Vorländer's work [5] goes beyond simple visualization to offer interactive immersive simulations. These approaches rely on binaural rendering, ambisonics, and geometric acoustic simulations. They serve as an important reference in this field, particularly for immersive architectural acoustic simulation. Physical phenomena thus become directly perceptible within the immersive experience: perception of reverberation, variations in thermal comfort, sound spatialization, and environmental sound propagation.

However, despite sensory immersion, these representations remain largely implicit: the user perceives the phenomenon but does not necessarily understand how it is characterized. In other words, hearing an acoustic difference does not mean understanding why it occurs, what parameters influence it, or how to act upon it. This distinction between perception and understanding appears to be a central obstacle in the literature.

Furthermore, the work of Kristin Nedlich, Arne Nykänen and Björn Hellström [18] introduces a particularly interesting dimension: 'sketching sound', where the user directly influences the sound environment within an interactive design framework.

However, despite their technical sophistication, these systems have several pedagogical limitations: complex interfaces, expert-oriented design, lack of cognitive guidance, and insufficient explanation of the phenomena. Most systems

implicitly assume that the user already possesses: an understanding of acoustics, an understanding of the parameters being manipulated, and the ability to interpret the perceptual effects observed.

Unlike metaphor-oriented approaches, these systems generally prioritize perceptual fidelity over interpretability. The objective is not necessarily to explain acoustics, but to reproduce it as accurately as possible. Several works extend these simulations toward architectural evaluation contexts. Llopis et al. [21] and Llorca-Bofi and Vorländer [15] use immersive auditory VR to evaluate architectural spaces before construction through precomputed acoustic simulations. Similarly, Guastamacchia et al. [16] validate VR systems for perceptual assessment of lecture room acoustics.

Other works explore immersive environmental acoustics at urban scale. Nykänen et al. [17] simulate environmental noise planning models inside VR, allowing users to experience urban soundscapes immersively.

Doggett et al. [18] take this a step further by investigating the impact of room acoustics on cognitive performance and well-being in immersive environments. Their work highlights that acoustics influence: concentration, stress, cognitive efficiency and subjective comfort.

A key feature of these simulation systems is that they rely on auditory realism rather than explicit perceptual mediation. The phenomenon is experienced directly through hearing itself, thereby limiting the understanding of physical acoustic phenomena.

- **Immersive metaphors of other phenomena**

Gan et al. [2], for example, combine BIM and virtual reality to visualise the aerodynamic behaviour around modular buildings (see Figure 4). Airflow and wind comfort are represented using: vector fields, animated particle trajectories, dynamic flow visualisations, and colour-coded intensity gradients. Here, the airflow is visually represented as movement within the space.

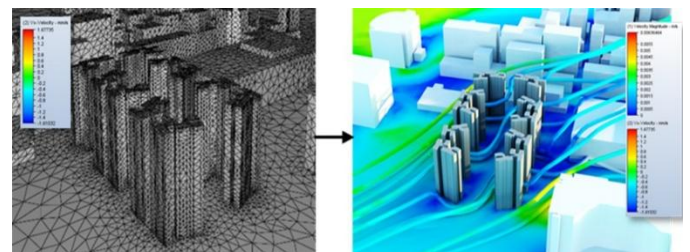


Fig. 4. Visualisation of airflow comfort around modular buildings [2]

In addition, Mehrzad Shahinmoghadam, Worawan Natephra and Ali Motamedi [3] use BIM and IoT (Internet of Things) data to represent thermal comfort in immersive environments using: heat maps, environmental overlays, and real-time thermal indicators.

This research is extremely valuable for the design of our education system, as it demonstrates: how invisible phenomena can be spatially externalised; how multimodal correspondences

can support perception; and how environmental data can be transformed into immersive experiences.

Although effective for experts, these representations still require users to decode symbolic conventions; they also reveal a limitation common to acoustic systems: very few approaches are designed with pedagogical intelligibility for novice learners in mind.

- **Immersive simulation of other phenomena**

Similarly, thermal comfort simulations [1, 3, 4, 19] aim to replicate environmental conditions within immersive spaces through: BIM integration, IoT sensors, real-time environmental monitoring, and the simulation of heating, ventilation and air-conditioning (HVAC) systems.

These systems enable users to: observe thermal variations, compare environmental configurations, assess comfort conditions and test building performance. Once again, this work allows the consequences of user interactions with the environment to be applied and compared, and validates this approach.

A. Focus synthesis for this study

- **Synthesis of some sensories methaphors**

TABLE I. SYNTHESIS OF SOME SENSORIES METHAPHORS

Room acoustics [7, 8]	
Type of metaphor	Visual acoustic particles (wavefronts, propagation fields, color gradients)
Results	High rendering fidelity Strong spatial exploration capabilities Helps users perceive propagation paths Spatial interactions
Main limitations	Limited for novices Strong cognitive load Requires interpretation expertise
Spatial acoustics [9]	
Type of metaphor	Visual-auditory perceptual correspondence
Results	Improves intuitive perception of spatial sound behaviour
Main limitations	Limited physical explanation Mostly perceptual rather than conceptual
Urban acoustic comfort [10]	
Type of metaphor	Symbolic color-coding on building façades
Results	Rapid comparison of acoustic performances between configurations Immediately readable
Main limitations	Relies on symbolic conventions Interpretation depends on prior knowledge
Aerodynamic airflow [2]	
Type of metaphor	Visual flow rays Symbolic color-coding on flow rays
Results	Strong perception of airflow dynamics and wind comfort
Main limitations	Scientific visualisations remain technically dense
Thermal comfort [3]	
Type of metaphor	Heat maps Environmental overlays Thermal indicators

Results	Makes environmental variations directly observable Facilitates comparison between conditions
Main limitations	Symbolic interpretation remains necessary

- **Synthesis of perceptual mappings**

Research on multisensory metaphors [20, 11, 12] explores the possibility of translating a physical phenomenon into another sensory modality, , which is called “perceptual mapping”. These approaches introduce several types of perceptual mapping:

- Analogical mappings, where the phenomenon retains a perceptual similarity to its representation (exemple: sound amplitude represented by light intensity, or sound propagation represented by animated waves);
- Symbolic mappings, where the representation is based on conventions (exemple: red = hot, and blue = cold);
- Arbitrary mappings, where the phenomenon/representation relationship does not have an immediate perceptual intuition and requires prior learning.

As synthesised in Table 2, the analyzed studies show that analogical mappings more effectively facilitate: intuitive understanding, memorization, and the construction of coherent mental models. In contrast, arbitrary mappings significantly increase: cognitive load and the risk of misinterpretation.

TABLE II. SYNTHESIS OF PERCEPTUAL MAPPINGS

Mapping method	Principle	Result
Analogical	The phenomenon retains a perceptual similarity to its representation	Enhance intuitive understanding, memorization, and the construction of coherent mental models
Symbolic	The representation is based on conventions	Enhance intuitive understanding, memorization, and the construction of coherent mental models
Arbitrary	Does not have an immediate perceptual intuition Requires prior learning	Increase cognitive load and the risk of misinterpretation

- **Perception, immersion and cognitive load**

Research focused on immersive perception [18, 14, 21, 22, 23, 12] shows that perceptual quality strongly influences the sense of presence, engagement, and immersion. Recent research on immersive multimodality [11, 12] also highlights the importance of perceptual correspondences between sensory modalities.

However, several studies show that: immersion does not automatically lead to better understanding; an immersive environment can cause cognitive overload; and multiple, unprioritized stimuli can divert attention away from educational objectives.

Furthermore, the work of Celia Hodent [24] provides an essential theoretical framework here. It shows that: user

attention is limited, information overload impairs learning, and interaction must be structured around explicit cognitive objectives. This issue becomes central in complex physical simulation environments.

- **Immersive learning environments**

Research on immersive learning [24, 23, 6] shows that VR environments offer significant potential for: engagement, experiential learning, active exploration and real-world application.

However, several recurring limitations emerge: cognitive overload, unguided interactions, difficulty in knowledge transfer, and poor pedagogical structure. The literature highlights that a highly realistic immersive environment can become difficult for novice learners to understand.

- **Professional training and transferability**

Research in the field of professional training [1, 3, 25, 4, 19, 6] shows a growing interest in: immersive BIM, IoT integration, real-time simulations, and contextualised building environments. These approaches bring learning much closer to real-world situations, environmental assessments and comfort-related issues.

However, few studies actually assess: the understanding of invisible phenomena and the transfer of skills outside VR. The majority of evaluations still focus primarily on: engagement, user satisfaction, perceived realism, and immediate performance.

- **General critics**

Overall, the literature shows that virtual reality offers great potential for representing invisible physical phenomena through immersive simulations and sensory metaphors. Existing research succeeds in externalising complex environmental data and enhancing the perceptual exploration of acoustic, thermal and aerodynamic behaviours.

However, several limitations remain insufficiently addressed in relation to the objectives of professional education: most systems are aimed at experts rather than novice learners; immersive realism is often prioritised at the expense of conceptual intelligibility; perceptual metaphors are rarely designed explicitly for educational purposes; cognitive accessibility remains poorly assessed; and few studies experimentally compare different configurations of sensory metaphors.

These observations highlight a significant research opportunity: to design immersive sensory metaphors specifically intended to facilitate the understanding of invisible acoustic phenomena in vocational construction education. The proposed project therefore aims to bridge the gap between the realism of immersive simulations and pedagogically interpretable representations, by developing interactive, multimodal acoustic metaphors adapted to beginner learners.

III. METHOD

Based on all our observations, and in order to meet the objectives, for our “pedagogical” tool, we propose to develop and test a solution aimed at addressing the following questions:

- RQ1: Can sensory metaphors in virtual reality improve the understanding of invisible acoustic phenomena in professional building education?
- RQ2: Which sensory metaphor configurations provide the best balance between conceptual understanding and cognitive load?

To address these questions, we propose and present in the following section of this article:

- A virtual reality learning environment dedicated to building acoustics education;
- A framework for designing sensory metaphors based on simplified acoustic simulation models;
- An experimental protocol comparing several metaphor configurations through knowledge assessments and cognitive load measurements conducted with high school students.

A. Visual Rendering of Acoustic Physical Phenomena

- **Representation of Acoustic Energy**



Fig. 5. Visual metaphor render

The proposed system models sound energy through an immersive analogical metaphor in which acoustic energy is represented as dynamic spherical entities propagating through space (see Figure 5 and Figure 6). The objective of this metaphor is to transform invisible acoustic phenomena into directly perceptible spatial behaviours while preserving intuitive perceptual correspondences between the physical phenomenon and its representation.

Each acoustic sphere is defined by a set of physical and perceptual parameters:

- Amplitude (dB) represented by sphere size;
- Frequency (Hz) represented through a logarithmic colour gradient (linear scale);
- Propagation direction represented through vectorial movement;
- Propagation velocity.

Each sphere represents a discrete unit of acoustic energy whose behaviour evolves according to simplified acoustic propagation rules. The simulation reproduces several physical acoustic phenomena:

- Sound propagation in space, represented through vectorial displacement of spheres according to propagation direction and velocity;
- Reverberation, represented through sphere reflections on surfaces and materials;
- Sound transmission, represented through the propagation of energy across materials and room boundaries;
- Sound attenuation, represented through a progressive reduction in sphere size corresponding to energy loss.

There will be several rooms where these acoustic phenomena (reverberation time and sound transmission) can also be experienced.

• Physical parameterisation

Material entities are associated with simplified acoustic properties allowing the simulation of frequency-dependent interactions (explicited below):

- Global absorption coefficient;
- Sound reduction index;
- Absorption coefficient at: 125 Hz, 250 Hz, 500 Hz, 1000 Hz, 2000 Hz, 4000 Hz, 8000 Hz.
- Surface roughness coefficient;
- Surface area;
- Air absorption coefficient.

This parameterisation enables the simulation of differentiated interactions between sound energy and architectural materials while preserving real-time execution constraints in immersive environments.

• Interaction algorithm

For each sphere, the following operations are performed:

- Vectorial displacement according to: propagation direction and propagation velocity.
- Collision detection with architectural materials.
- Computation of acoustic interactions.

When a collision between a sphere and an acoustic material occurs (see Figure 6), the sphere splits. Technically, it (the parent sphere: A_0 in the figure 6) makes its child sphere (from the next generation: A_{1tr} in the figure 6) visible and teleports it to the same coordinates as itself. It is then reflected in the room, whilst the other (its child sphere: A_{1r} in the figure 6) passes through the material. two phenomena appears : the reflexion and the transmission. For both of these phenomena, the properties of reflected and transmitted spheres (amplitude and visual size) change depending on the physical properties of the material they encounter, in accordance with the principles of acoustics [26] and European standards [27].

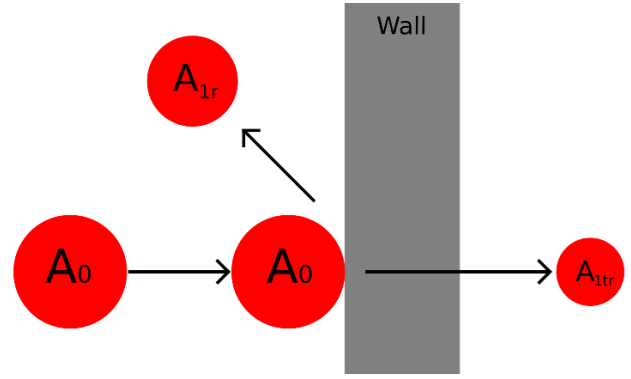


Fig. 6. Diagram showing the division of the acoustic wave into energy transmitted to another room and reflected energy

B. Haptic Metaphors

In addition to visual metaphors, the system introduces analogical haptic metaphors intended to render acoustic properties physically perceptible through vibration feedback in VR controllers.

• Reverberation Time

When the user initiates the metaphor, the system computes the reverberation time of the room in which the sound source is located, based on the acoustic properties of the materials in that room. If the user is in the same room, they will feel a vibration of maximum intensity lasting for a duration equal to the previously calculated reverberation time. This metaphor therefore works in a delayed time frame.

Here is a proposed formula for sound absorption when a sphere comes into contact with an acoustic material:

$$A_1 = A_0 * (1 - \alpha W) \quad (1)$$

Where A_1 is the reflected amplitude in decibels (dB), A_0 is the incident amplitude in decibels (dB), and αW is the absorption coefficient, between 0 to 1. In fact, the absorption coefficient, which appears in material data sheets, represents the proportion of sound energy absorbed by the material during the reflexion on a scale of 0 to 1.

• Sound transmission

When the user initiates the metaphor, the system start to computes the sound transmission rate (in decibels) on each frames between the sound source and the position of the VR controller, based on the acoustic materials located between the two and the sound amplitude emitted by the sound source. The calculated transmission value is rendered by a continuous haptic feedback signal, the intensity of which is proportional to the previously calculated transmission rate. This metaphor therefore works in real time.

Here is a proposed formula for sound transmission when a sphere pass through an acoustic material:

$$A_1 = A_0 - R W \quad (2)$$

Where A_I is the transmitted amplitude in decibels (dB), A_0 is the incident amplitude in decibels (dB), and R_w is the sound reduction index, between 0 to 1. In fact, the sound reduction index, which appears in material data sheets, represents the proportion of sound energy absorbed by the material during the transmission on a scale of 0 to 1.

C. User Interactions

The device that users employ relies solely on the Meta Quest 2 VR headset with controllers. No additional equipment is required.

- **Sound source and metaphors parametrisation**

The left-hand VR controller allows users to manipulate the parameters of the metaphors (see Figure 7, left panel).



Fig. 7. 3D sound source, metaphors panel (the left panel) and material panels (the right panel)

With this interface, user can (top to bottom):

- Modify source parameters: sound power (dB), frequency (Hz);
- Enable or disable the sound emission (defined by the sources parameters), characterized by the activated metaphors with the “Play” button;
- Interact with visual metaphor parameters: speed, freeze time functionality, number of spheres per generation, number of generations ;
- Interact with haptic metaphor parameters: “reverberation time” or “transmission” modes, apply the norm (the controllers vibrate only above the norm), dampen vibration coefficient.

- **Material parametrisation**

The right-hand VR controller provides contextual information about the acoustic properties of the material the user is interacting with. Here, the user can switch between materials using the arrows on either side of the material name (see Figure 7, right panel).

It is important to note that changes to the material, the parameters of the metaphors, or the sound source have a real-time impact on the visual metaphors of reflection and

transmission, as well as on the haptic metaphor of transmission. However, they have a pre-calculated impact on the haptic metaphor of reflection. Indeed, as mentioned earlier, the reverberation time is calculated at the moment the sound is emitted.

In addition, the user can move around virtually by teleporting and turn their virtual body in small increments to allow them to interact with the materials in the room. They can also move their virtual reality headsets and controllers through virtual objects, although they cannot move to another room.

D. Experimental Protocol

This experimental study was conducted with eight students in the building field with acoustic physics knowledges at a professional high school, eight high school students without acoustic physics knowledges at a general high school, and sixteen students in the building field without acoustic physics knowledges at a professional high school (see Table 3). The aim of the study was to assess whether immersive sensory metaphors could facilitate the understanding of invisible acoustic phenomena whilst remaining cognitively accessible and usable for novice and advanced learners.

TABLE III. SYNTHESIS OF CHOSEN PARTICIPANTS

Public	Field	Prior knowledge of acoustics	Distribution by experimental conditions
Vauban (Nice)	Building	Yes	EVM: 4 EHM: 1 EVHM: 3
Avesnières (Laval)	General scientific	No	EVM: 3 EHM: 3 EVHM: 2
François Hennebique (Liévin)	Building	No	EVM: 6 EHM: 5 EVHM: 5

Participants interacted with the system individually using a virtual reality setup comprising a virtual reality headset and handheld VR controllers. To reduce physical fatigue and minimise the risk of motion sickness, participants remained seated throughout the experiment. They were, however, able to explore the virtual environment freely using teleportation and instantaneous rotations.

The protocol was designed to directly address the research questions introduced previously:

- RQ1: Can sensory metaphors in virtual reality improve the understanding of invisible acoustic phenomena in professional building education?
- RQ2: Which sensory metaphor configurations provide the best balance between conceptual understanding and cognitive load?

The experimental session followed five successive stages. Participants first completed a unique-choice questionnaire (QCU pre-test) designed to evaluate their initial knowledge of building acoustics concepts, including: sound propagation, acoustic reverberation, acoustic transmission, material absorption properties. Secondly, each participants were then assigned to one experimental condition:

- EVM: Experimental Visual Metaphors only;
- EHM: Experimental Haptic Metaphors only;
- EVHM: Experimental Visual and Haptic Metaphors.

The greyed-out settings (see Figure 7) are not available under the experimental conditions.. This session was deliberately designed to be exploratory and interactive, although each participant was given the same set of instructions: “activate the metaphor(s)”, “adjust the sound source parameters (frequency, amplitude) and reactivate the metaphor(s)”, “interact with a partition, change the material of the partition and reactivate the metaphor(s)”, “repeat these steps as you wish to understand the acoustic mechanisms”. Participants were informed that they could stop the virtual reality session at any time if they felt uncomfortable, tired or experienced symptoms of motion sickness. Although no strict minimum duration was imposed, the session was deliberately stopped after ten minutes in order to limit prolonged exposure and ensure comparable experimental conditions across participants.

Thirdly, immediately after the virtual reality session, participants completed a second single-choice questionnaire (post-test SCQ) that was identical in every respect to the pre-test SCQ. The aim here was to ascertain whether the VR tool and exploratory learning enable professional high school students to get useful concepts.

Finally, participants then completed two standardized subjective questionnaires: the Raw Task Load Index (RTLX), used to assess perceived cognitive workload during the immersive experience, and the System Usability Scale (SUS), used to evaluate the perceived usability and accessibility of the VR system. Participants were able to answer open-ended questions about the tool’s strengths, limitations and areas for improvement. These qualitative feedbacks were optional.

IV. RESULTS AND DISCUSSION

One participant exhibited significant symptoms of cyberillness during the immersive session and was therefore excluded from the analysis in order to preserve the validity of the data collected. Furthermore, at this stage, before conducting a larger-scale experiment, only thirty-two participants split between three high schools completed the entire experimental protocol, and their profiles are not comparable. Consequently, the statistical relevance of the results remains limited. This study should therefore be regarded as a preliminary exploratory assessment rather than a large-scale validation study.

A. Knowledge and Time Performance

• Knowledge score evolution

The analysis of the pre- and post-test QCU results (see Figure 8) highlights contrasted tendencies between both the three experimental conditions and the participants’ prior knowledge of acoustic physics. The objective of this evaluation was to measure both the evolution of conceptual understanding of acoustic phenomena and the time required to answer the questionnaire after the immersive experience.

Students from Vauban High School, enrolled in professional building education and already familiar with acoustic concepts,

achieved the highest pre-test scores (7.5/12 on average) (see figure 8). Their post-test performance remained globally stable (7.25/12), with a slight decrease (-0.25 points). This result suggests a ceiling effect: participants already possessed relatively well-established mental models before the VR session, limiting observable learning gains.

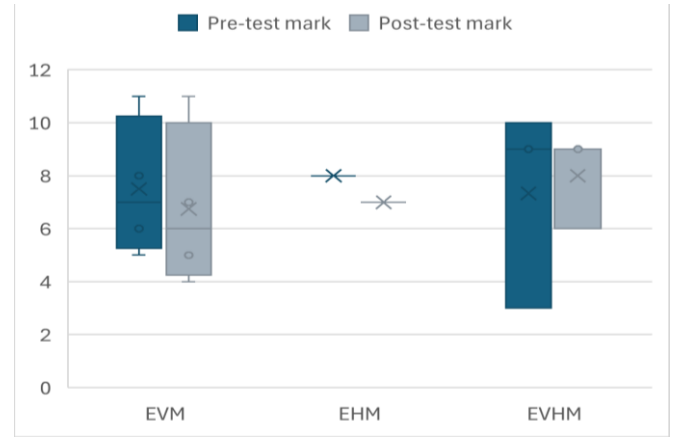


Fig. 8. QCU mark performance between pre-test QCU and post-test QCU with Vauban’s public

Students from Avesnières High School, enrolled in a general scientific field but without prior acoustics education, obtained a lower initial score (4.75/12) (see figure 9). The overall post-test score remained unchanged. However, substantial differences appeared between conditions. The haptic condition (EHM) showed the strongest improvement (3.33/12 to 6/12), whereas the visual-haptic condition (EVHM) decreased slightly. These results suggest that learners with a scientific background may be able to infer physical relationships from isolated sensory cues, although the small sample size prevents definitive conclusions.

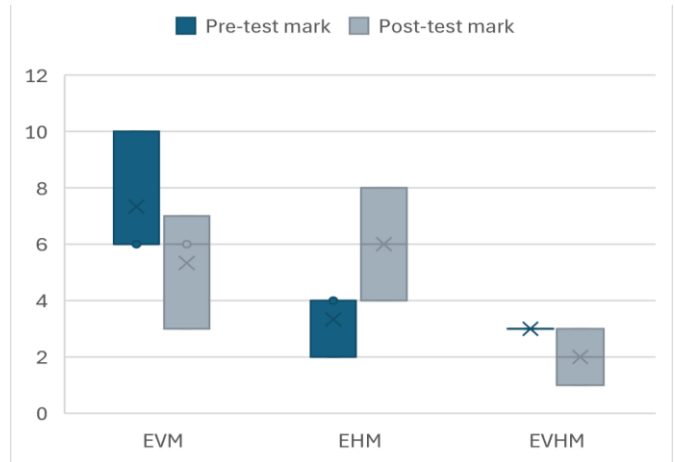


Fig. 9. QCU mark performance between pre-test QCU and post-test QCU with Avesnières’ public

Students from François Hennebique High School, enrolled in vocational building education but without prior acoustics knowledge, achieved a comparable pre-test score (4.5/12) (see figure 10). Their average score increased slightly to 4.62/12 after the experiment. The largest improvement again appeared in the haptic condition (4.6/12 to 5.8/12), while the visual and multimodal conditions remained stable or decreased slightly.

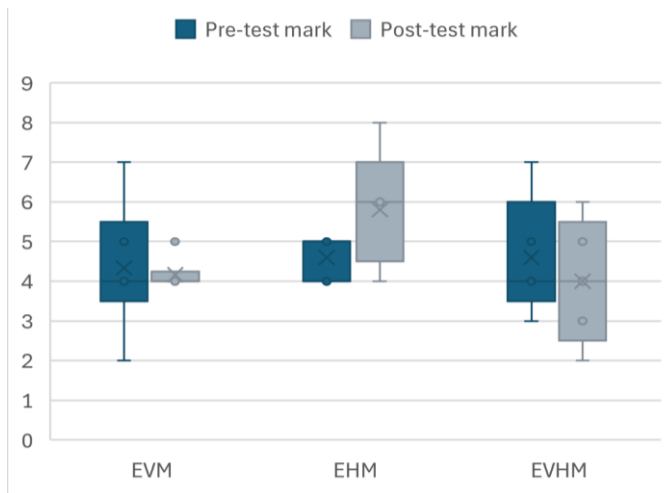


Fig. 10. QCU mark performance between pre-test QCU and post-test QCU with François Hannebique's public

Across the three schools, learning gains remain relatively limited. The normalized gains range from 3.54% to 19.17%, indicating that a single short VR session is insufficient to produce substantial conceptual acquisition. Nevertheless, the results suggest that sensory metaphors may contribute to stabilizing or slightly improving learners' mental representations, particularly among participants without prior acoustics expertise.

Overall, the visual-haptic condition remains the most promising configuration in terms of consistency, while the haptic condition unexpectedly produced several of the largest individual improvements among novice learners.

• Questionnaire completion time

For the Vauban group, completion time increased substantially after the immersive session (+3 min 45 s on average) (see figure 11). This increase may indicate deeper reflection and a more analytical interpretation of the questions following exposure to the metaphors. Participants may have reconsidered concepts they previously answered automatically based on prior knowledge.

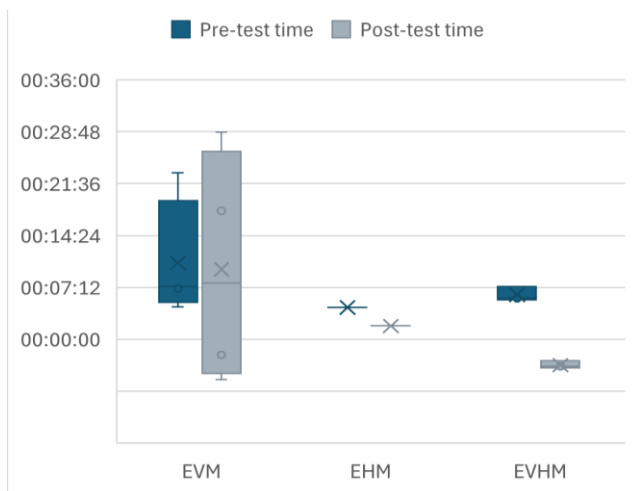


Fig. 11. QCU time performance between pre-test QCU and post-test QCU with Vauban's public

In contrast, the François Hannebique group showed a strong reduction in completion time (-3 min 25 s on average) (see figure 12), while maintaining relatively stable scores. This result suggests that the immersive experience may have facilitated faster access to mental representations of acoustic phenomena without significantly altering overall knowledge levels.

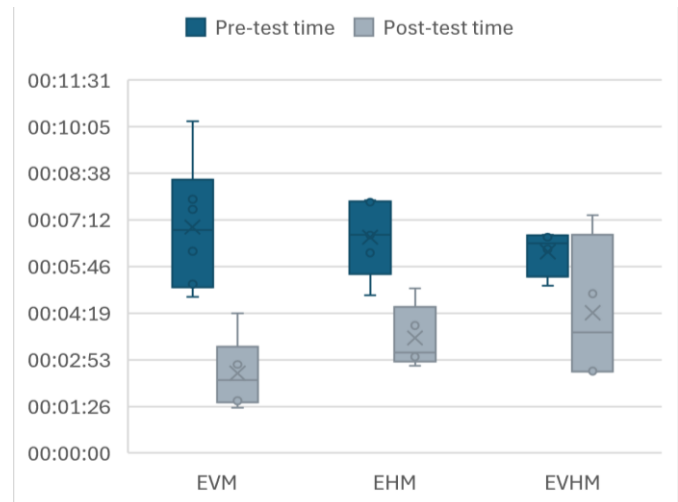


Fig. 12. QCU time performance between pre-test QCU and post-test QCU with François Hannebique's public

The Avesnières group exhibited almost no change (+10 s) (see figure 13), suggesting that the VR experience neither substantially accelerated nor slowed conceptual reasoning.

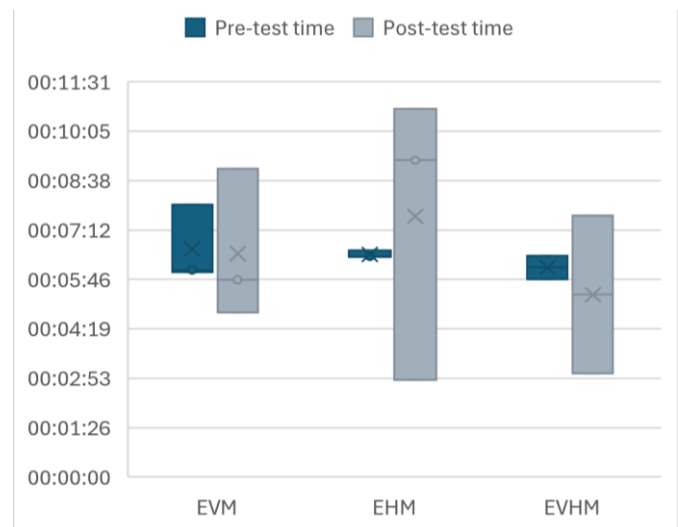


Fig. 13. QCU time performance between pre-test QCU and post-test QCU with Avesnières' public

Interestingly, the shortest post-test completion times were generally observed in the multimodal condition (EVHM), particularly for novice participants. This tendency supports the hypothesis that combining visual and haptic metaphors may facilitate more direct interpretation of acoustic propagation mechanisms.

Although these observations remain exploratory, they suggest that sensory metaphors may influence not only learning outcomes but also the cognitive strategies mobilized during problem solving.

B. Raw Task Load Index score

The RTLX questionnaire was used to evaluate perceived cognitive workload during the immersive experience. The overall cognitive load remained moderate across all three studies, with average scores ranging from 46.67 (with Vauban's public) to 51.25 (with François Hannebique's public) out of 100.

The Vauban group reported the lowest workload in the visual condition (41.67) and the highest workload in the haptic condition (73.33) (see Figure 14). This result may be explained by participants' existing acoustic knowledge, which allowed them to interpret visual representations more easily while making isolated haptic cues appear ambiguous.

In contrast, the Avesnières (see Figure 15) and François Hannebique (see Figure 16) groups displayed much smaller differences between conditions. Cognitive workload remained relatively balanced across visual, haptic and multimodal metaphors, with values generally between 40 and 55 points.

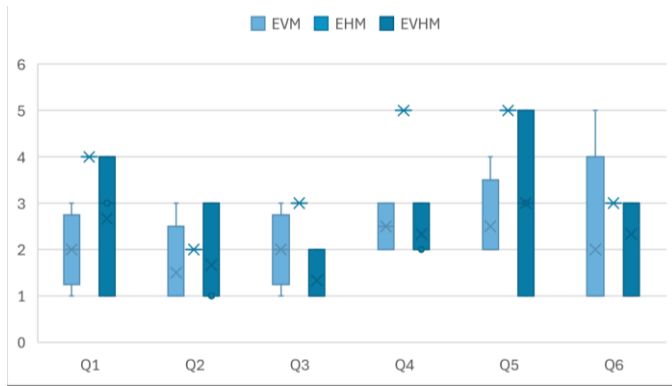


Fig. 14. Raw Task Load index scores for each experimental condition on each question with Vauban's public

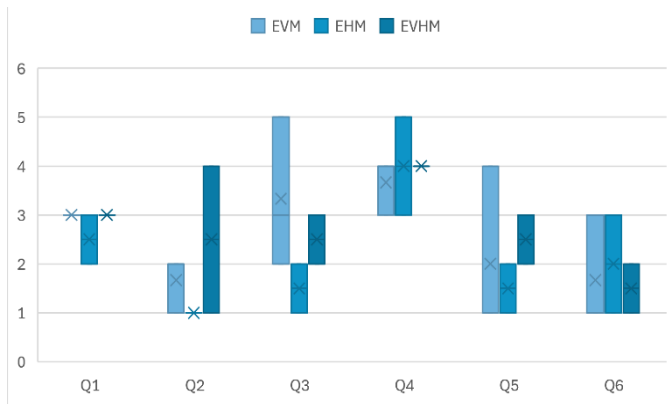


Fig. 15. Raw Task Load index scores for each experimental condition on each question with Avesnières' public

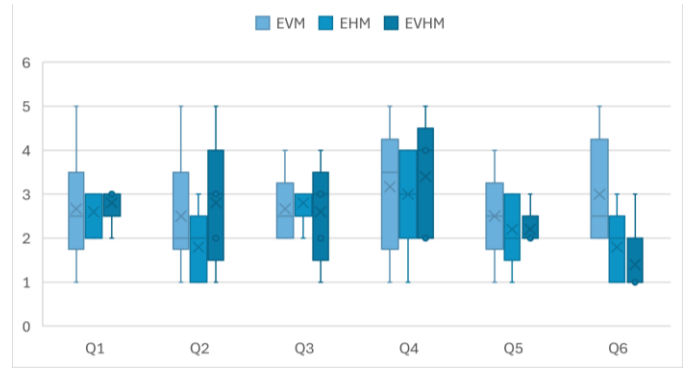


Fig. 16. Raw Task Load index scores for each experimental condition on each question with François Hannebique's public

A notable finding is that the multimodal condition (EVHM) never produced the highest workload score in any of the three schools. This observation suggests that combining sensory channels does not necessarily increase cognitive demand. Instead, complementary sensory information may facilitate interpretation and reduce uncertainty.

These findings are consistent with previous research on multimodal perception, which argues that coherent sensory correspondences can improve information processing without generating excessive mental workload.

Although the number of participants was small, these findings are consistent with the literature on perceptual correspondences and multimodal immersion, which suggests that consistent sensory correspondences can facilitate comprehension without increasing cognitive load.

C. System Usability Scale score

The SUS questionnaire was used to evaluate the perceived usability of the VR learning environment. The overall SUS score increased from 58.75 (with Vauban's public) (see Figure 17) to 75.62 (with Avesnières' public) (see Figure 18) and 62.36 (with François Hannebique's public) (see Figure 19). This result indicates that the system remains functional and understandable, but still requires significant usability improvements before broader educational deployment.

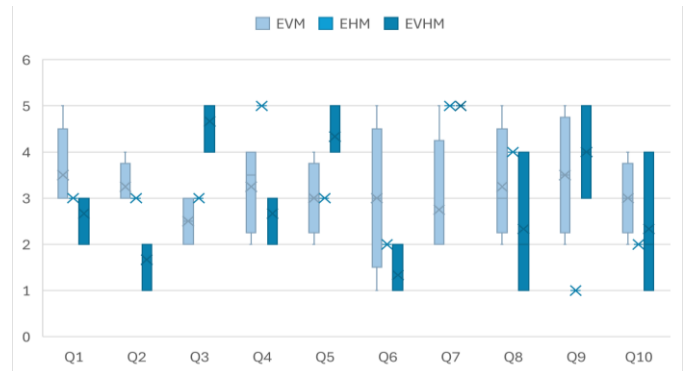


Fig. 17. System Usability Scale score for each experimental condition on each question with Vauban's public

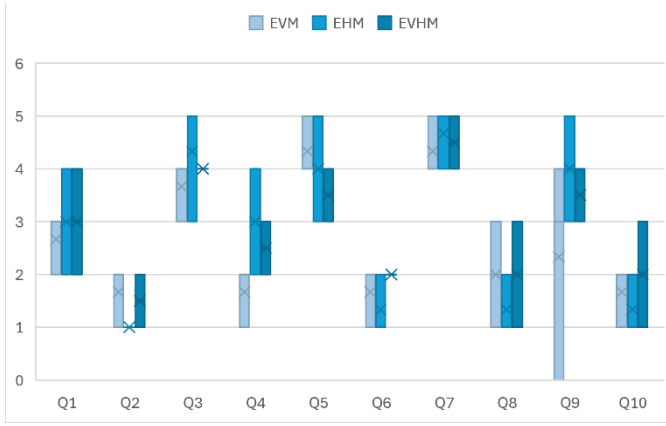


Fig. 18. System Usability Scale score for each experimental condition on each question with Avesnières' public

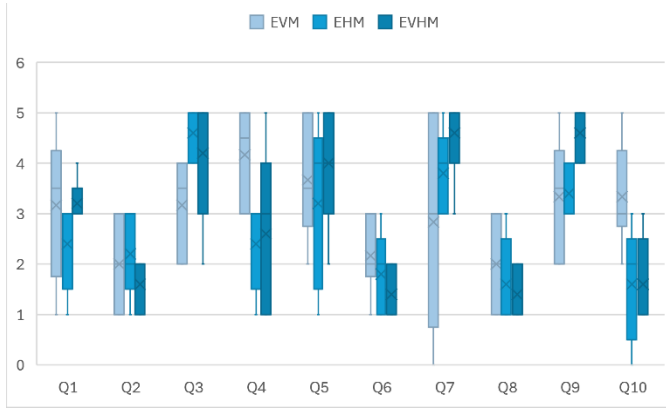


Fig. 19. System Usability Scale score for each experimental condition on each question with François Hannebique's public

The highest usability scores were consistently observed in the multimodal condition (EVHM), reaching 75.83, 71.25 and 80.00 respectively. These values correspond to good usability according to standard SUS interpretation scales. The visual-only condition (EVM) produced more heterogeneous evaluations, ranging from 48.75 to 74.17. Similarly, the haptic-only condition ranged from 47.50 to 80.00.

These results suggest that the combination of visual and haptic metaphors provides a more coherent and understandable user experience than either modality alone. Participants generally reported that multimodal feedback facilitated the interpretation of acoustic interactions and improved the perceived consistency of the simulation.

Interestingly, usability remained relatively high even among participants with no prior acoustics knowledge, indicating that the proposed interaction model can be understood without extensive domain expertise. Overall, the SUS results strongly support the pedagogical relevance of multimodal sensory metaphors in immersive educational environments.

V. CONCLUSION

This study investigated the use of sensory metaphors in virtual reality to support the understanding of invisible acoustic phenomena in building education. Three experimental

conditions were compared: visual metaphors (EVM), haptic metaphors (EHM), and combined visual-haptic metaphors (EVHM), across three distinct learner populations differing in educational background and prior acoustics knowledge.

The results indicate that learning gains remain relatively modest after a single short VR session. However, several encouraging tendencies emerge. First, participants without prior acoustics knowledge generally benefited more from the immersive experience than those already familiar with the subject. Second, multimodal sensory metaphors consistently achieved the highest usability scores while maintaining moderate cognitive workload. Third, reductions in questionnaire completion times observed among novice professional learners suggest that immersive representations may facilitate the construction of operational mental models.

Several limitations remain. Sample sizes within each condition were relatively small, preventing robust statistical analyses. The study also relied on short-term assessments and did not investigate long-term retention or transfer of learning to real professional situations. Furthermore, the three learner populations differed not only in acoustics knowledge but also in educational background, introducing potential confounding factors. Furthermore, one participant experienced cybersickness during the experiment and was excluded from the analysis. Although participants were free to stop the VR session at any moment, this highlights the importance of comfort and cybersickness management in immersive educational environments.

Despite these limitations, the experiment provides several important perspectives for future work. The results suggest that: multimodal sensory metaphors may improve the interpretability of invisible physical phenomena, perceptual coherence between modalities appears essential for usability and understanding, immersive educational environments should prioritize pedagogical intelligibility rather than perceptual realism alone.

Future work will therefore focus on: increasing the participant number, reduce the cognitive load, improving interaction usability, and evaluating long-term learning transfer in building education contexts.

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